PYTHON CODING CHALLENGE

Customer Data Cleaning System

Topic: Python String Cleaning Functions | Level: Beginner | Suggested time: 20 minutes

1. Business Scenario

You are working as a Data Analyst in an e-commerce company. The company receives customer information from different sources. Because of inconsistent data entry, the information contains extra spaces, inconsistent uppercase/lowercase letters, and unwanted spaces or hyphens in phone numbers.

Your task is to write a Python program that cleans the customer information before it is stored in the company's database.

2. Input Data

```
customer_name = "  aNIL   kumAR  "
email = "  ANIL.KUMAR@GMAIL.COM "
city = "  hyDERABAD  "
phone = "  +91-98765  43210 "
feedback = "  GOOD   product,   fast delivery!  "
```

3. Your Tasks

1. Remove unwanted spaces from the beginning and end of the customer name.
2. Convert the customer name into proper title case and reduce multiple internal spaces to one.
3. Convert the email address into lowercase and remove leading/trailing spaces.
4. Convert the city name into proper title case and remove extra spaces.
5. Remove spaces and hyphens from the phone number.
6. Clean multiple spaces in the feedback while preserving the words and punctuation.
7. Display all cleaned customer details in a neat format.

4. Expected Output

```
----- CLEANED CUSTOMER DETAILS -----

Customer Name : Anil Kumar
Email         : anil.kumar@gmail.com
City          : Hyderabad
Phone         : +919876543210
Feedback      : GOOD product, fast delivery!
```

Think like a Data Analyst: clean the data accurately, then explain why you selected each string method.